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identifying or determining the nature of disease while treatment involves the curative aspects in order to eliminate the disease causing agent. Clinical laboratory tests help a physician in correct diagnosis and treatment, coupled with advanced communication technology, telemedicine, an emerging area are vital especially in rural health care.

12.1 Diagnostic and Monitoring Instruments

Stethoscope

Stethoscope is a medical device used to hear the internal sounds of the human body such as heart beat, sounds due to inhalation and exhalation of air in the lungs, stomach, intestinal movements and also foetal movements. The modern electronic stethoscope is a high precision instrument used to hear with clarity the internal sounds of the human body even in a noisy environment or through the heavy clothing of the patient. A typical stethoscope consists of a disc shaped resonator that is placed against chest with two rubber tubes connected to earpieces. This is simply known as binaural stethoscope. It is a simple but an effective diagnostic tool to localize problems and to diagnose diseases (Figure12.1).

Figure12.1 Clinical stethoscope and electronic stethoscope

Clinical significance of stethoscope

1. Stethoscope helps to find the normal and abnormal heart beat sounds and also to diagnose valve functions.
2. It helps to diagnose lung diseases such as pneumonia, pulmonary edema, bronchitis and pleuritis.
3. Stethoscopes along with sphygmomanometer are used to read the blood pressure.
4. It outlines the status of cardiac, respiratory and intestinal disorders.

Sphygmomanometer

Sphygmomanometer is a device that measures the blood pressure hence known as blood pressure meter, blood pressure monitor or blood pressure gauge. It consists of a rubber cuff which is wrapped around the upper arm and the other end is connected to a mercury manometer which directly measures the pressure in a uniformly graduated scale ((Figure12.2).

By gradually increasing and reducing the pressure in the cuff, the systolic and diastolic blood pressure is clearly monitored by the sounds of arterial blood flow. The cuff is smoothly wrapped around the upper arm 3 cm above the elbow. The resonator of the stethoscope is placed on the brachial artery at the elbow to hear the blood flow sounds. The cuff is then rapidly inflated to 180 mm Hg and is then slowly deflated. As the pressure in the cuff falls a “whooshing” sound is heard then the blood flow starts again in the artery. This indicates the patient’s systolic pressure. Then the pressure is slowly released until no sound is heard. This is measured as the diastolic